



PLTW DIGITAL ELECTRONICS · PORTFOLIO ITEM · 15 POINTS

Soldering Projects: 1.1.8 & 1.1.9

Jitterbug Vibration-Motor Bot & Random Number Generator Board

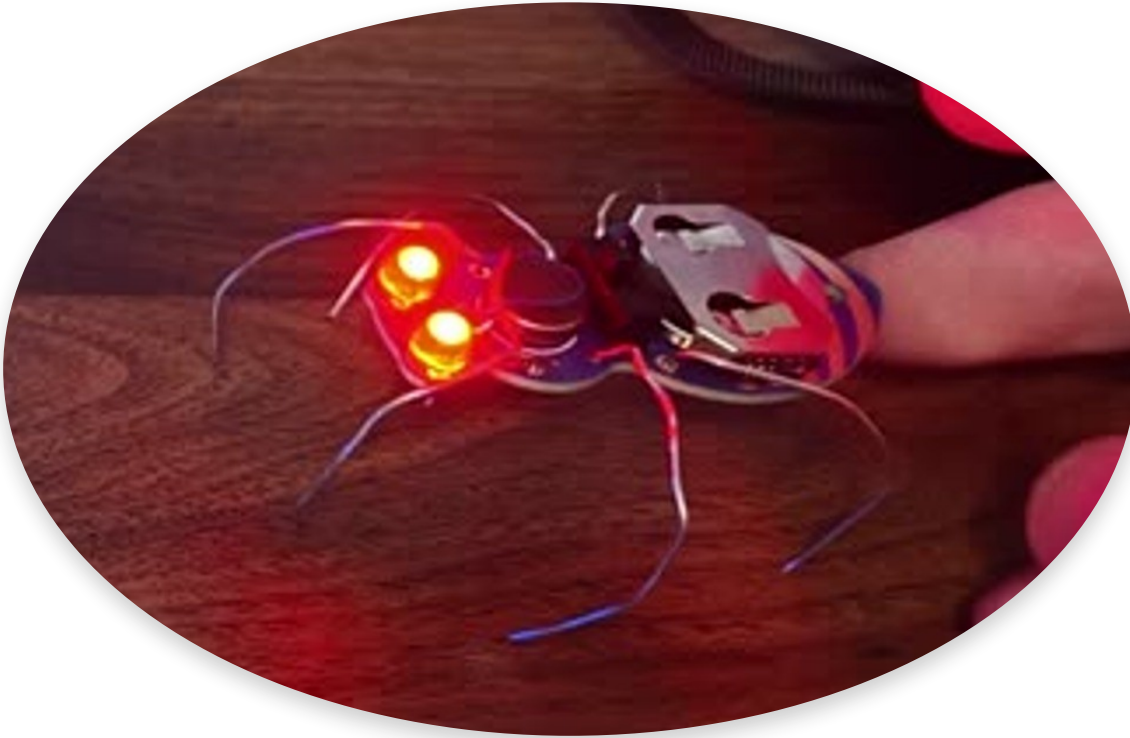


Conner Glover

ACTIVITY 1.1.8

Jitterbug — Vibration-Motor Robot

Learn To Solder Kit, soldered from a bare PCB



Finished Jitterbug — battery installed, both LED eyes lit

Kit: Learn To Solder Kits “Jitterbug” vibration-motor bot

Power: 3 V, non-rechargeable CR2032 coin cell through an SPDT slide switch

Circuit: Motor and both LED eyes wired in parallel off the switched supply, sharing a common return to battery negative

Mechanical: 3 bent-wire legs soldered in for support (no electrical function)

Build time: ~30 minutes to fully populate the board

Jitterbug — Right / Wrong / Learned

R Right

Soldered the entire board — motor, both LED eyes, battery holder, switch, and legs — in about 30 minutes.

Circuit worked the first time the battery was inserted: eyes lit and the motor ran.

W Wrong

The vibration motor's lead snapped off its solder joint after the first successful test.

Installed a replacement motor — one of its leads snapped off the same way before the fix held.

L Learned

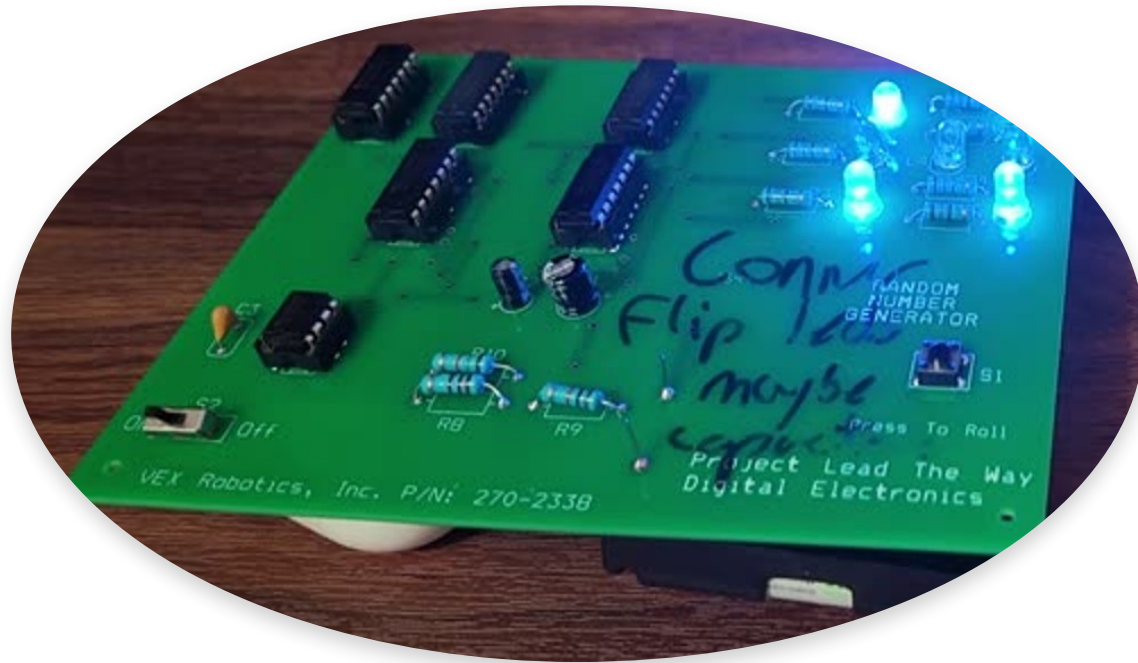
The motor leads are very thin, fragile wire that won't tolerate flexing after they're soldered down.

Dressing the lead flat against the board before soldering (instead of leaving slack to flex) is what finally kept the joint from breaking again.

ACTIVITY 1.1.9

Random Number Generator Board

PLTW / VEX Robotics kit, P/N 270-2338



Finished RNG board, powered on — press-button LEDs lit

Kit: PLTW / VEX Robotics “Random Number Generator” board (P/N 270-2338)

Function: Press-button (“Press To Roll”) triggers logic that lights one LED at random

Components: Logic ICs, resistor network, and LED array hand-soldered to the bare board

Build time: ~55 minutes (one class period) to fully populate the board

Random Number Generator — Right / Wrong / Learned



Right

Soldered every joint on the board — ICs, resistors, and the full LED array — in a single ~55-minute class period.



Wrong

Every LED on the board was soldered in backwards (anode/cathode reversed), so none of them lit.

Fixing it took two more class periods: one to desolder and pull all the old LEDs, one to reseat new LEDs in the correct orientation.



Learned

Desoldering already-placed LEDs stresses the package — the L6 LED didn't survive the rework and still shows no output.

Checking each LED's long lead (anode) against the schematic before soldering is far cheaper than a full desolder-and-replace pass afterward.

Modeling the RNG's Logic in Multisim

Decode: 74LS04 / 74LS08 / 74LS32 (NOT / AND / OR) gates decode the flip-flop outputs to light exactly one of LED1–LED7 per cycle — the same “pick one at random” behavior as the soldered board's push-button roll.